Xiang (Shawn) Zhang

E-mail: xiangzhg@umich.edu

https://shawnzhg.github.io/ | Google Scholar | Github

EDUCATION

University of Michigan, Ann Arbor United States

Major: M. S. in Data Science *08/2025-06/2027*

Average GPA: **3.9**

Core Courses: Machine learning (A), Action and Perception, Data Privacy, Web System.

Capital University of Economics and Business Beijing, China

Major: B. S. in Data Science *09/2021-07/2025*

Average GPA: **4.0**

Core Courses: Natural language processing, Operational research, Advanced Algebra, Artificial Intelligence, Data Mining.

RESEARCH INTERESTS

Representation learning, post-training, and agents for large language models, about how models learn, reason, and act, and how we can tell when they have actually improved.

PUBLICATIONS & PREPRINTS (* equal contribution)

- Han, Z.*, **Zhang, X.* (co-first author)**, Han, B., Liu, K., Hu, D., & Liu, J. KV-Skill: Forging Expertise in the Model's Native Language. arXiv:2608.05475. Under review at **ACL Rolling Review** (Oct 2026 cycle).
  Follow-up work: Li & Chen (HKUST; under review at ICLR 2027, arXiv:2610.06950) build directly on KV-Skill as their central point of comparison, adopting its operator design, benchmarks, and released checkpoint
- Han, Z.*, **Zhang, X.* (co-first author)**, Guan, L.*, Hu, D., Liu, K., Chang, K., & Liu, J. What Can a Leaderboard Certify? Compositional Controllability for Fair Evaluation and Training of Biomedical Literature-Review Agents. arXiv:2609.32318. Under review at **ICLR 2027**.

RESEARCH EXPERIENCE

LLM Post-training, Agents, Knowledge Graph and Latent Reasoning Ann Arbor, USA

Research Assistant *at University of Michigan Liu lab (Jie Liu)* *01/2026-Present*

- **KV-Skill (externalized task knowledge for LLMs): Proposed the core idea of representing reusable task skills** as external factorized associative operators that a frozen LLM can load on demand, and developed the RL formulation (**RL-from-skill and RL-from-scratch)**. Shaped the final design through extensive ablations and led the training and experimental evaluation. On Qwen3.5-4B LiveMath, the same skill scores 52.0 as text and 77.2 as a KV-Skill, **outperformed LoRA**, prefix tuning, and soft-prefix baselines in 7/8 matched-budget settings, and enabled multiple skills to coexist without forgetting. Integrated into PanKGraph (pankgraph.org) for Cypher generation during question answering.
- **SCRIBE** (**RL post-training for long-horizon agents**): **Designed** and built the **SCRIBE agent, reward formulation, and training pipeline for long-horizon** bio literature review. To address weak **credit assignment** from trajectory level GRPO rewards, introduced rubric-grounded, multi-dimensional rewards at stage boundaries and translated the project's compositional-controllability theory into trainable supervision over intermediate agent decisions. On the same literature-review evaluation, **SCRIBE outperformed Claude Science, Claude Code, OpenAI Responses API, and Elicit,** and is deployed as the Literature Writing workflow in GLKB (glkb.org).
- Biomedical **Deep research agent (reliable literature synthesis): Designed** the **Investigate** workflow in GLKB.org, a research agent harness for reducing hallucination and citation fabrication in biomedical literature deep research, combining compress-before-reason retrieval, verbatim-grounded full-text answering, and a mechanical faithfulness gate. Improved **fact-level completeness from 0.442 to 0.645** and **entity coverage from 0.676 to 0.823,** while reducing inference cost by 55% with zero fabricated citations.
- **Latent-space reasoning (efficient reasoning under RL): Investigated** whether redundant CoT computation could be selectively moved into latent space to reduce KV-cache while remaining amenable to RL post-training. **Developed a redundancy-based routing signal** that separated filler steps offline (AUC ≈ 0.72), but found that it provided almost no

usable RL signal, the learned policy skipped only 0.1% of steps. Further identified that removing the discrete token bottleneck impaired error correction, with a single latent step reducing accuracy from 64.5% to 55.0% on Qwen2.5 1.5B.

Reasoning Language Models (Post training) Beijing, China / Remote

Research Assistant *at Institute of Automation, Chinese Academy of Sciences(CASIA)* *06/2025- 12/2025*

- Depth-Quality Erosion **(diagnosing GRPO training): Identified** that rising aggregate Pass@1 during GRPO can mask systematic regressions across reasoning depths. On DeepSeek-R1-Distill-Qwen-1.5B, analyzed training at per-depth granularity and uncovered a Simpson's-paradox-like mode-seeking failure in which pass rate fell across individual depth bins by up to 26%, even while aggregate Pass@1 increased by 2.0 pp; validated the phenomenon across MATH-500, GSM8K, and AIME.
- **Structured credit assignment for GRPO:** Motivated by the failure above, **proposed a two-level credit-assignment framework addressing both missing learning signal across prompts and diluted credit across tokens.** STAGE-GRPO stratifies advantages using online per-prompt difficulty estimation, preserving informative gradients even for hard prompts whose rollout groups are predominantly incorrect; prefix-contrastive credit assignment then localizes this signal to decision-critical tokens by contrasting divergence points between same-prefix correct and incorrect rollouts, rather than broadcasting one terminal advantage uniformly across the sequence. STAGE-GRPO eliminated the observed depth-wise degradation and improved MATH-500 Pass@1 to 0.868 (+4.2 pp vs. base; +2.2 pp vs. GRPO).

Spatiotemporal Drought Forecasting via Vision Transformer Beijing, China

Research Assistant *at Chinese Academy of Agricultural Sciences(CAAS)* *02/2025- 06/2025*

- **Spatiotemporal drought forecasting:** Designed a forecasting framework over 24 biweekly frames of multivariate climate fields to predict future gridded SPI/SPEI across multiple accumulation windows and lead times, explicitly separating genuine forecast skill from persistence and partially observed target windows.
- **Latent climate dynamics**: Proposed a climate state encoder to latent dynamics to probabilistic decoder architecture for 1 and 6 month forecasting. Pretrained a ViT style encoder on decades of unlabeled ERA5 fields via masked reconstruction, then used a weight-tied monthly latent transition model conditioned on land–ocean and seasonal-forecast signals to roll climate states forward before decoding SPEI maps and drought probabilities; designed multi-step training to limit rollout error and scaled experiments with multi node, multi GPU training.

OTHER PUBLICATIONS (* equal contribution)

- Liu, E.*, **Zhang, X.***, Pan, P.*, Wang, Y., Liu, B., Zhang, L., & Mei, X. Agricultural technology gap affects climate change adaptation and mitigation based on patent data. npj Climate and Atmospheric Science, under review.
- Zhang, L., **Zhang, X.**, Gao, J., Lei, S., Yan, C., & Liu, E. An insight into the technological landscape and competitive situation of the biodegradable PBAT industry based on multidimensional patent analysis. IEEE Transactions on Engineering Management, under review.

SKILLS

- Distributed Training: **Deepspeed**, **DDP**, **FSDP**; RL infra: **TRL/veRL/OpenRLHF**; **vLLM**; **SGLang**
- Proficient in Python (PyTorch, Pandas, NumPy), Java, R, SQL, Claude code

INTERNSHIP

Baidu.com Times Technology (Beijing) Co., Ltd. Beijing, China

Algorithm Intern *07/2024-10/2024*

- Developed and optimized a speech recognition system using Transformer architecture to achieve high-precision audio-to-text transcription, particularly for dialect data
- Built Python-based web scraping tools to collect video data, extracted audio features, and used the features to improve speech-to-text accuracy through a customized encoder structure
- Applied dynamic time warping to align ASR and OCR outputs, improving text alignment accuracy for multi-language and dialect audio

HONORS AND AWARDS

- First-Class Merit-based Scholarship (Top 5%) 2023-2024
- First Prize, National College Student Market Research and Analysis Competition 2024
- Scientific Innovation Scholarship 2024
- Project Leader, Beijing University Student Innovation & Entrepreneurship Development Program 2024